

THE PHYSICAL AI SYSTEMS SCOPE: THREE-PILLAR CONVERGENCE

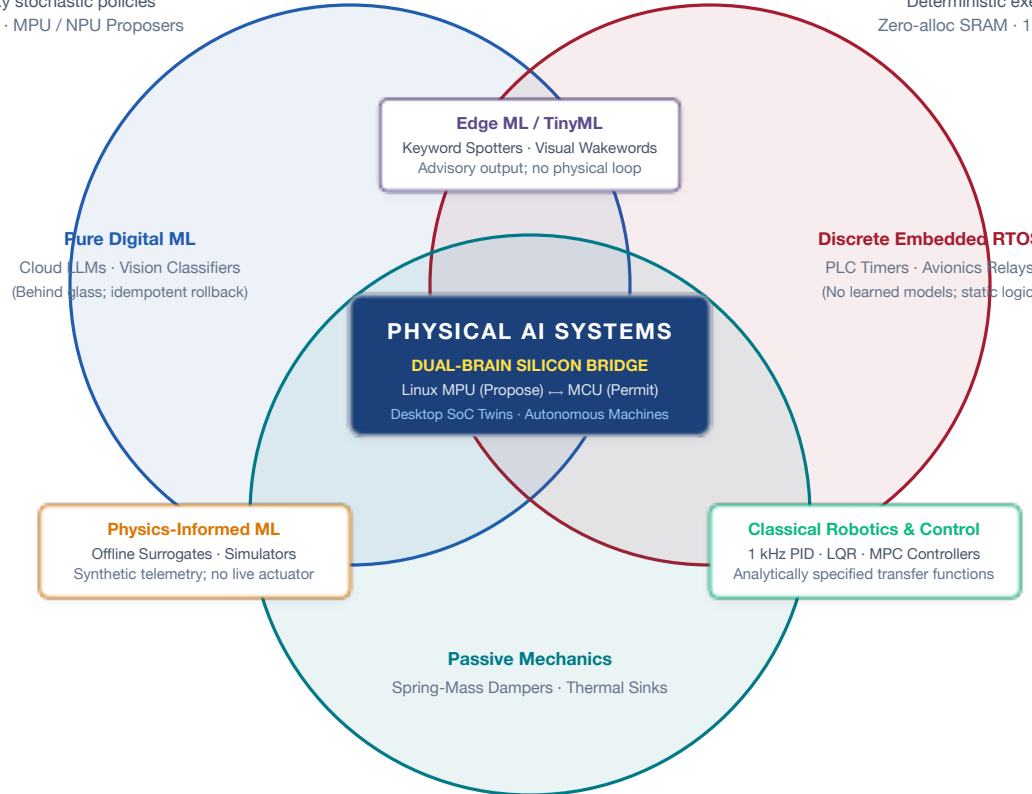
An ML Systems Perspective: High-Capacity Learned Models · Real-Time Embedded Silicon · Physical Dynamics

MACHINE LEARNING SYSTEMS

High-capacity stochastic policies
VLMs · Diffusion · MPU / NPU Proposers

REAL-TIME EMBEDDED SYSTEMS

Deterministic execution & WCET
Zero-alloc SRAM · 1 kHz MCU Referees



PHYSICAL MECHANICS & THERMODYNAMICS

Inertia ($p=mv$) · Non-reversible energy ($I^2 R$) · Closed causal loop ($s_{t+1} \sim P(s|s_t, a_t)$)